221ECS004	COMPUTATIONAL INTELLIGENCE	CATEGORY	L	T	P	CREDIT
		Program Elective 1	3	0	0	3

Preamble: The aim of this course is to provide the students with the knowledge and skills required to design and implement effective and efficient Computational Intelligence solutions to problems for which a direct solution is impractical or unknown. This course covers concepts of fuzzy logic, genetic algorithms, and swarm optimization techniques. The learners will be able to provide Fuzzy and AI –based solutions to real world problems.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Apply fuzzy logic to handle uncertainty and solve engineering problems. **(Cognitive Knowledge Level: Apply)**
CO 2	Apply Fuzzy Logic Inference methods in building intelligent machines. **(Cognitive Knowledge Level: Apply)**
CO 3	Design genetic algorithms for optimized solutions in engineering problems. **(Cognitive Knowledge Level: Analyze)**
CO 4	Analyze the problem scenarios and apply Ant colony system to solve real optimization problems. **(Cognitive Knowledge Level: Analyze)**
CO 5	Apply PSO algorithm to solve real world problems. **(Cognitive Knowledge Level: Apply)**
CO6	Design, develop and implement solutions based on computational intelligence concepts and techniques. **(Cognitive Knowledge Level: Create)**

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and developmentwork in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1				✔		✔	
CO 2	✔		✔	✔	✔	✔	
CO 3	✔		✔	✔	✔	✔	
CO 4	✔		✔	✔	✔	✔	
CO 5	✔		✔	✔	✔	✔	
CO 6	✔	✔	✔	✔	✔	✔	✔

Assessment Pattern

Bloom's Category	**End Semester Examination**
Apply	70%-80%
Analyze	30%-40%
Evaluate	
Create	

Mark distribution

Total Marks	**CIE**	**ESE**	**ESE Duration**
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation: 40 marks

i. Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks

ii. Course based task / Seminar/ Data collection and interpretation : 15 marks

iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.
Course based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is 40+20 = 60 %.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Let V = {A, B, C, D} be the set of four kinds of vitamins, F = {f_1, f_2, f_3} be three kinds of fruits containing the vitamins to various extents, and D = {d_1, d_2, d_3} be the set of three diseases that are caused by deficiency of these vitamins. Vitamin contents of the fruits are expressed with the help of the fuzzy relation R over F x V, and the extent of which diseases are caused the deficiency of these vitamins is given by the fuzzy relation S over V x D. Relations R and S are given below

$$R = [0.5\ 0.2 \quad 0.2\ 0.7 \quad 0.4\ 0.4 \quad 0.1\ 0.1\ 0.4\ 0.3\ 0.8\ 0.1\ ]S$$
$$= [0.3\ 0.5\ 0.1\ 0.8\ \ 0.7\ 0.4\ \ 0.9\ 0.1\ 0.5\ 0.5\ \ 0.2\ 0.3\ ]$$

Find the correlation between the amount of certain fruit that should be taken while suffering from a disease.

Course Outcome 2 (CO2):

1. In mechanics, the energy of a moving body is called kinetic energy. Suppose we model mass and velocity as inputs to a moving body and energy as output. Observe the system for a while and the following rule is deduced.

IF x is small and y is high

THEN z is medium

The graphical representation of rule is given below. Let the inputs given are 0.35kg and 55m/s. What will the output using Mamdani inference? Any defuzzification method can be used to obtain the crisp single output.

Course Outcome 3(CO3):

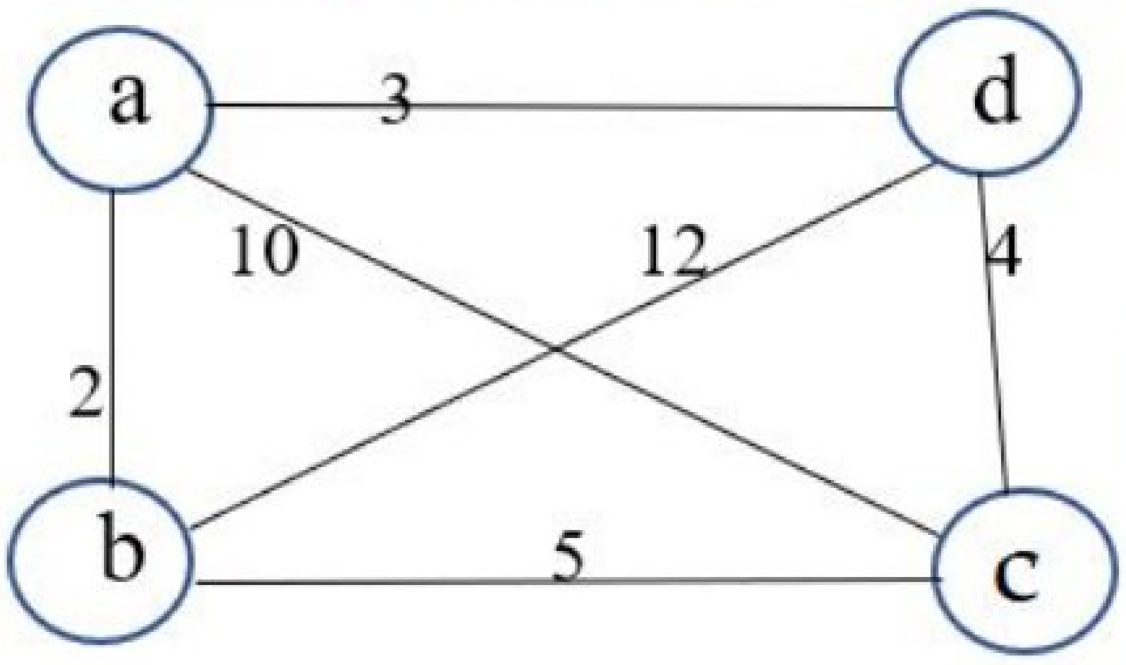
1. Describe how Roulette wheel is used for selection. Draw the Roulette wheel for six chromosomes corresponding to the table given below.

Chromosome #	***Fitness***
1	10
2	5
3	25
4	15
5	30
6	20

Course Outcome 4 (CO4):

1. Consider an Ant Colony System based on Ant Quantity model for solving the following Travelling Salesman Problem. Compute the pheromone content at each of the edges after 4 steps(1 iteration). Assume pheromone decay factor ρ=0.1, Q = 120. Assume initial pheromone of 50 units at each of the edges and that three ants k1, k2 and k3 follow the paths given below in the first iteration.

k1= a b c d a; k2=a c b d a; k3=a d c b a



2. Six jobs go first on machine A, then on machine B, and finally on machine C. The order of the completion of the jobs in the three machines is given in Table

Jobs	Processing time(hr)		
	Machine A	Machine B	Machine C
1	8	3	8
2	3	4	7
3	7	5	6
4	2	2	9
5	5	1	10
6	1	6	9

Find the sequence of jobs that minimizes the time required to complete the jobs using the ACS model.

Course Outcome 5 (CO5):

1. Consider a particle swarm optimization system composed of three particles and maximum velocity 10. Assume that both the random numbers r1 and r2 used for computing the movement of the particle towards the individual best position and social best position are 0.5. Also assume that the space of solutions is the two-dimensional real valued space and the current state of swarm is as follows:

   Position of particles: x1 = (4,4); x2 = (8,3); x3 = (6,7)

   Individual best positions: x14,4) = *); x2* = (7,3); x3* = (5,6)

   Velocities: v1 = (2,2); v2 = (3,3); v3 = (4,4)

   What would be the next position of each particle after one iteration of the PSO algorithm if the inertia parameter ω that is used along with current velocity update formula is 0.8 ?

Course Outcome 6 (CO6):

1. Implement travelling salesman problem using appropriate optimization technique.

Model Question Paper

QP CODE:

Reg No: _______________

Name: __________________ **PAGES: 5**

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M. TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221ECS004

Course Name: Computational Intelligence

Max. Marks: 60 **Duration: 2.5 Hours**

PART A

Answer All Questions. Each Question Carries 5 Marks

1. Consider the set of Colours A= {Blue, Red, Orange, Yellow, Green}, Attributes B = {Bright, Warmth, Dullness}, Feelings C= {Unpleasant, happiness, Angry}. Given R and S where R is the relationship between colours and their attributes and S is the relationship between colour attributes and feelings created. Find the relationship Q between colours and feelings created **(5)**

R	Bright	Warmth	Dullness
Blue	0.8	0.6	0.4
Red	0.8	0.8	0.2
Orange	0.5	0.7	0.2
Yellow	0.3	0.6	0.5
Green	0.8	0.6	0.4

S	Unpleasant	Happiness	Angry
Bright	0.2	0.8	0.6
Warmth	0.4	0.7	0.8
Dullness	0.8	0.3	0.6

2. Develop a membership function for "Tall". Based on that devise membership function for "Very Tall". Explain how it is done **(5)**

3. Mention the importance of objective (fitness) function in genetic algorithm **(5)**

4. Describe how pheromone is updated. What is elitist / elastic ants ? Are they useful in this scenario? **(5)**

5. What is the significance of pbest and gbest particles in solving problems with particle swarm optimization? **(5)**

Part B

(Answer any five questions. Each question carries 7 marks)

6. (a) Consider the set of fruits F = {Apple, Orange, Lemon, Strawberry, Pineapple}. **(3)**

Let sweet fruits B= $\left\{\frac{0.8}{Apple} + \frac{0.6}{Orange} + \frac{0.2}{Lemon} + \frac{0.4}{Strawberry} + \frac{0.7}{Pineapple}\right\}$ and

Sour Fruits F= $\left\{\frac{0.6}{Apple} + \frac{08}{Orange} + \frac{0.9}{Lemon} + \frac{0.7}{Strawberry} + \frac{0.5}{Pineapple}\right\}$

Find Fruits that are Sweet or Sour, Sweet but not Sour, Sweet and Sour

(b) Consider two fuzzy Sets given by **(4)**

$$P = \left\{\frac{0.9}{short} + \frac{0.3}{medium} + \frac{0.5}{tall}\right\}$$

$$Q = \left\{\frac{0.7}{positive} + \frac{0.4}{zero} + \frac{0.8}{negative}\right\}$$

Find the fuzzy relation for the Cartesian product of P and Q i.e, R=P x Q .

Introduce a fuzzy set T given by

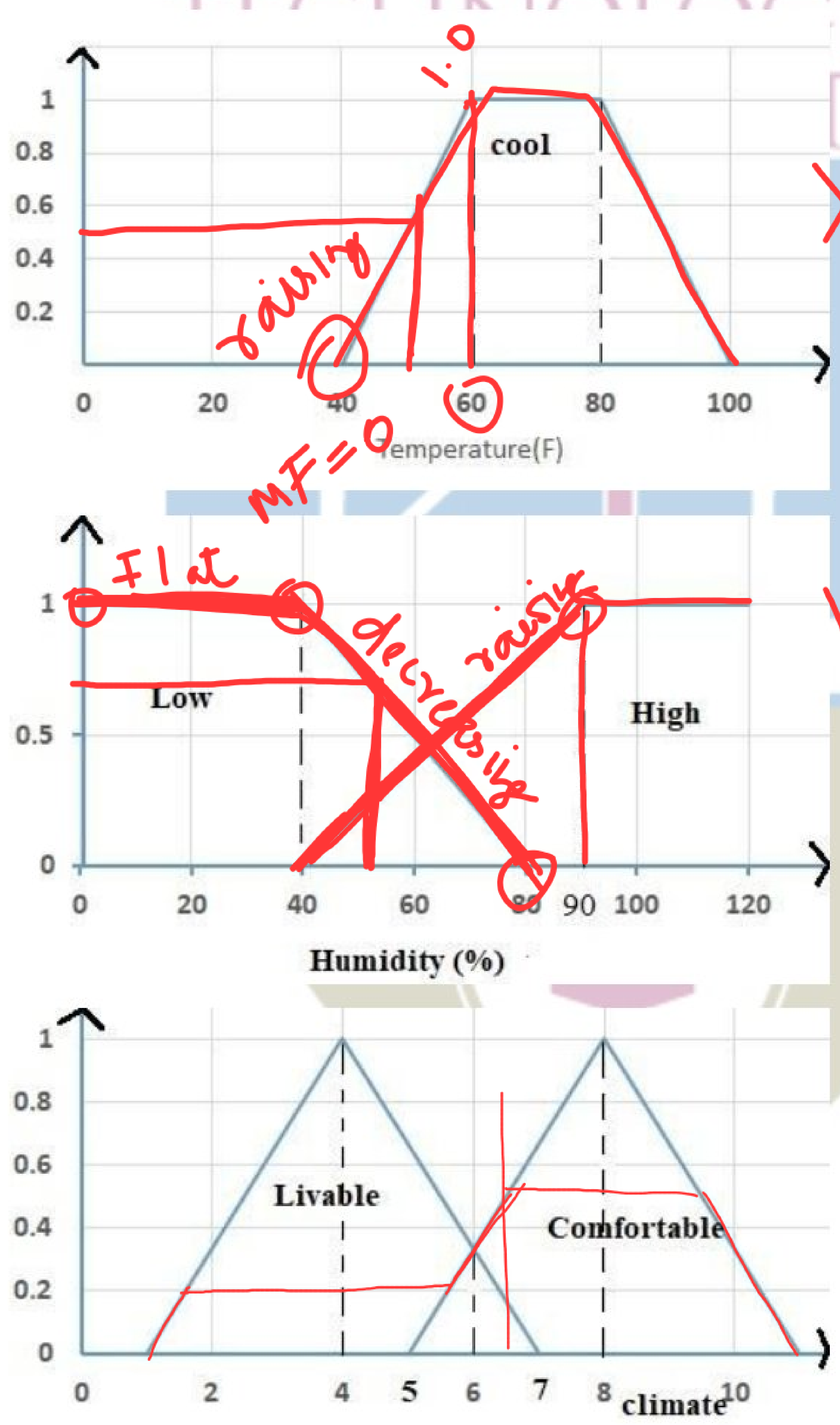
$$T = \left\{\frac{0.9}{short} + \frac{0.3}{medium} + \frac{0.6}{tall}\right\}$$

and Find T o R using max-min composition

7. Consider a Fuzzy Inference System for checking climate comfortability of human beings for long time living. The system accepts two inputs – temperature and humidity. The rules and membership functions of FIS is given below. Using Mamdani inference and center of sum, calculate output when the temperature is 50Fahrenheit and humidity is 50%. **(7)**

Rule 1: IF temperature is cool and humidity is low, THEN climate is comfortable.

Rule 2: IF temperature is cool and humidity is high, THEN climate is livable



The fuzzy sets "Easy Question Paper" and their corresponding "Student Performance" are given below

$$\text{Easy\_QP} = \left\{ \frac{0.8}{1} + \frac{0.2}{2} + \frac{0.6}{3} + \frac{0.7}{4} \right\}$$

$$\text{Stud\_Perf} = \left\{ \frac{0.3}{a} + \frac{0.4}{b} + \frac{0.8}{c} + \frac{0.9}{d} \frac{0.8}{1} + \frac{0.2}{2} + \frac{0.6}{3} + \frac{0.8}{4} + \frac{0.7}{5} \right\}$$

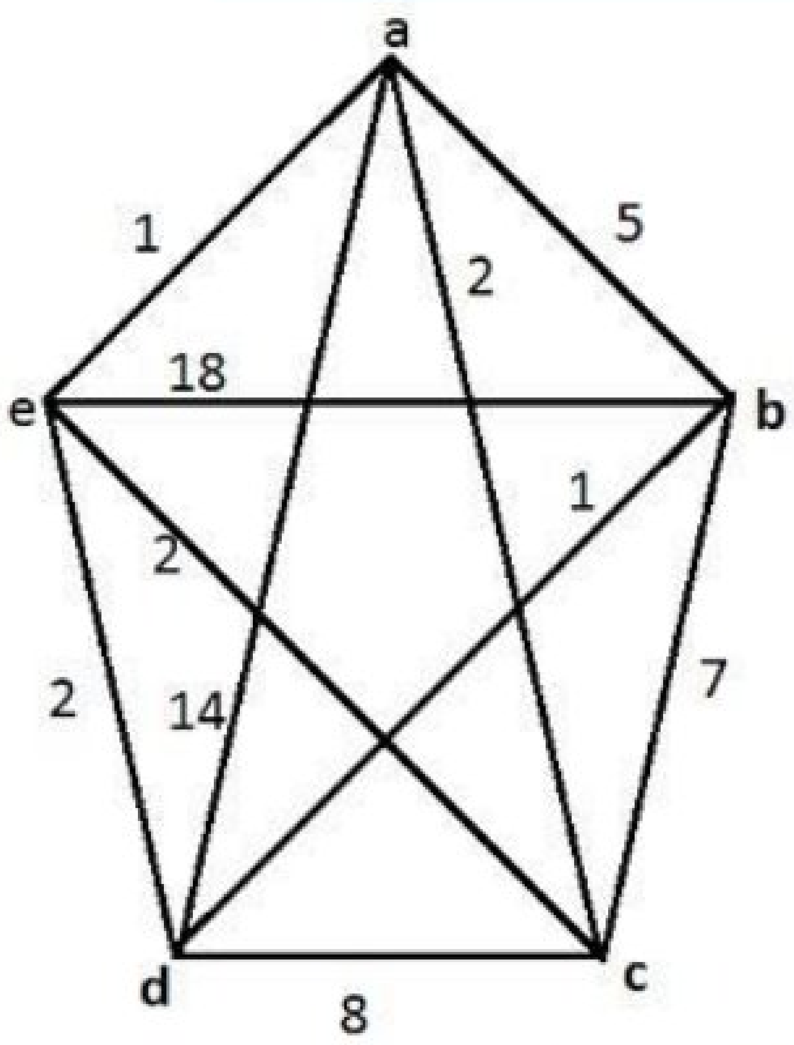
Find the performance of students c and d for the question paper "Somewhat Easy"

$$\text{Somewhat\_Easy} = \left\{ \frac{0.7}{1} + \frac{0.3}{2} + \frac{0.5}{3} + \frac{0.6}{4} \right\}$$

8. Explain any procedure to map a solution to the corresponding chromosome and vice versa in genetic algorithms. Also illustrate it with an example **(7)**

9. Describe two methods used to select individuals from a population for the mating pool in Genetic Algorithms **(7)**

10. (a) Consider the TSP with the following edge costs. Given the evaporation factor $\rho = 0.02$ and initial pheromone at all edges $T_{ij}=100$ **(1)**



What is the cost of best tour?

(b) Using the equation $T_{ij}(t+1)=(1-\rho)T_{ij}(t) + \Delta T_{ij}(t,t+1)$, compute the T_{ij} of the edge<a,c> when 10 ants uses the edges <a, c>, using the following models: **(6)**

i. Ant Density Model (Constant Q=10)

ii. Ant Quantity Model(Constant Q=100)

where Q is the constant related to the pheromone updation.

11. Describe Ant Colony System. What are the different types of Ant systems? **(7)**

12. Consider a particle swarm optimization system composed of three particles and maximum velocity 10. Assume that both the random numbers r1 and r2 used for computing the movement of the particle towards the individual best position and social best position are 0.5. Also assume that the space of solutions is the two-dimensional real valued space and the current state of swarm is as follows: **(7)**

Position of particles: x1 = (4,4); x2 = (8,3); x3 = (6,7)

Individual best positions: x1* = (4,4); x2* = (7,3); x3* = (5,6)

Velocities: v1 = (2,2); v2 = (3,3); v3 = (4,4)

What would be the next position of each particle after one iteration of the PSO algorithm if the inertia parameter ω that is used along with current velocity update formula is 0.8?

Syllabus

Module 1: Fuzzy Logic

Crisp sets vs fuzzy sets- Operations and properties of Fuzzy sets. Membership functions - Linguistic variables. Operations on fuzzy sets- Fuzzy laws- Operations on fuzzy relations, Fuzzy composition- Max- min**,** Max – product. Alpha-cut representation.

Module 2: Fuzzy Systems

Fuzzy Reasoning – GMP and GMT. Fuzzy Inference System: Defuzzification methods - Fuzzy Controllers -Mamdani FIS, Larsen Model

Module 3: Genetic Algorithms

Introduction to Genetic Algorithms – Theoretical foundation - GA encoding, decoding - GA operations – Elitism – GA parameters – Convergence. Multi-objective Genetic Algorithm – Pareto Ranking.

Module 4: Ant Colony Systems

Swarm intelligent systems - Background Ant colony systems – Biological systems- Development of the ant colony system- - Working - Pheromone updating- Types of ant systems- ACO algorithms for TSP

Module 5: Particle Swarm Optimization

Basic Model - Global Best PSO- Local Best PSO- Comparison of 'gbest' to 'lbest'- PSO Algorithm Parameters- Problem Formulation of PSO algorithm- Working. Rate of convergence improvements -Velocity clamping- Inertia weight- Constriction Coefficient- Boundary Conditions- Guaranteed Convergence PSO- Initialization, Stopping Criteria, Iteration Terms and Function Evaluation.

Course Plan

No	Topic	No. of Lectures (40)
1	**Module 1: Fuzzy Logic**	**9**
1.1	Crisp sets vs fuzzy sets, Operations and properties of Fuzzy sets	1
1.2	Membership functions	1
1.3	Linguistic Variables	1
1.4	Operations on fuzzy sets	1
1.5	Fuzzy laws	1
1.6	Operations on fuzzy relations	1
1.7	Fuzzy Composition- Max- min	1
1.8	Fuzzy Composition – Max- Product	1
1.9	Alpha-cut representation	1
2	**Module 2: Fuzzy Systems**	**7**
2.1	Fuzzy Reasoning – GMP	1
2.2	Fuzzy Reasoning –GMT	1
2.3	Fuzzy Inference System	1
2.4	Defuzzification methods	1
2.5	Fuzzy Controllers	1
2.6	Mamdani Model	1
2.7	Larsen Model	1

3	**Module 3: Genetic Algorithms**	**7**
3.1	Introduction to Genetic algorithm	1
3.2	Theoretical foundation	1
3.3	GA encoding - decoding	1
3.4	GA operations	1
3.5	Elitism, GA parameters, Convergence of GA	1
3.6	Multi – objective Genetic Algorithm	1
3.7	Pareto Ranking	1
4	**Module 4: Ant Colony Systems**	**8**
4.1	Swarm intelligent systems	1
4.2	Background	1
4.3	Ant colony systems – biological systems	1
4.4	Development of the ant colony system	1
4.5	Working	1
4.6	Pheromone updating	1
4.7	Types of ant systems	1
4.8	ACO algorithms for TSP	1
5	**Module 5: Particle Swarm Optimization**	**9**
5.1	Basic Model	1
5.2	Global Best PSO	1
5.3	Local Best PSO, Comparison of 'gbest' to 'lbest'	1
5.4	PSO Algorithm Parameters	1
5.5	Problem Formulation	1
5.6	Working	1
5.7	Rate of convergence improvements – velocity clamping	1
5.8	Inertia-weight - Constriction Coefficient- Boundary Conditions	1
5.9	Initialization, Stopping Criteria, Iteration Terms and Function Evaluation	1

References

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3. Xin-She Yang School of Science and Technology, Middlesex University London, Nature-Inspired Optimization Algorithms, Elsevier, First edition, 2014
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